

Affan Ahmed Khan

Full-Stack Developer | Real-Time Systems Engineer

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PROFESSIONAL SUMMARY

Full-Stack Developer specializing in real-time, scalable web applications and developer tooling. Experienced in building high-performance systems using React, Node.js, and WebSockets, with a strong focus on backend architecture, event-driven design, and performance optimization. Proven ability to design and ship production-grade SaaS platforms with efficient state management, secure authentication systems, and polished, responsive user interfaces.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, Java, C++, C#
Frontend: React, TypeScript, Tailwind CSS, Framer Motion, Zustand, TanStack Query
Backend: Node.js, Express, Prisma, PostgreSQL, REST APIs, Socket.io
Systems: Redis, BullMQ, AWS S3, Firebase, rrweb, Chrome Extension APIs
Data: SQL, Pandas, NumPy, Matplotlib, Seaborn
Architecture: Real-Time Systems, RBAC, JWT Auth, Event-Driven Design

EXPERIENCE

Federal Board of Revenue (FBR)

Jul 2025 – Sep 2025

Intern

- Processed and validated large-scale taxation datasets, ensuring consistency across structured financial records and reports.
- Analyzed tabular data to identify discrepancies and supported internal reporting workflows used in compliance processes.
- Collaborated with administrative teams to streamline record-handling procedures and reduce manual inefficiencies.

PROJECTS

NeoPlane — Real-Time Communication Platform

React, Node.js, Socket.io, Prisma, PostgreSQL, AWS S3

- Engineered a low-latency messaging system using Socket.io with efficient event broadcasting and connection handling.
- Designed scalable backend (Express + Prisma) with JWT authentication and optimized relational queries.
- Built media pipeline with AWS S3 + Sharp for upload, compression, and retrieval workflows.
- Optimized client performance using Zustand + TanStack Query, reducing redundant API calls.

Nexus — Enterprise Task Management Platform

React, TypeScript, Node.js, PostgreSQL, Prisma, Socket.io

- Developed real-time Kanban system with live updates, presence tracking, and activity streams.
- Implemented RBAC with JWT + refresh tokens for secure multi-user environments.
- Leveraged React Query + Zustand for optimized state management with optimistic UI updates.
- Built analytics dashboards and scalable features including pagination and audit logs.

E-Commerce Engine & Analytics Platform

TypeScript, Node.js, PostgreSQL, Redis, BullMQ

- Architected high-throughput REST API handling complex relational business data.
- Implemented async job pipelines using Redis + BullMQ for automation workflows.
- Designed analytics engine computing KPIs using optimized queries and snapshot strategies.

Universal Web Debugger (Chrome Extension)

TypeScript, Chrome APIs, Shadow DOM, rrweb

- Built browser extension for runtime audits (performance, accessibility, CSS conflicts).
- Used Shadow DOM for UI isolation and safe DOM injection without style interference.
- Integrated rrweb session replay for debugging complex user interaction flows.

EDUCATION

NUST — Military College of Signals (MCS), Islamabad

Expected 2027

BS Software Engineering CGPA: 3.07

- Relevant Coursework:** Machine Learning, Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Computer Networks, Database Systems